Physiologically Guided R-Centered Deep Learning for ECG Wave Delineation Across Heterogeneous Databases

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Electrocardiogram (ECG) wave delineation is fundamental for automated cardiac analysis, enabling accurate identification of clinically significant waveform boundaries. Although deep learning methods have achieved promising performance, they typically formulate delineation as a sequence labeling task without explicitly incorporating the physiological structure of the cardiac cycle, limiting generalization across heterogeneous datasets. We propose a physiologically guided R-centered CNN–BiLSTM framework that combines deterministic R-peak localization with neural delineation of P and T waves while preserving physiologically meaningful temporal relationships. A systematic experimental study investigates the effects of loss formulations, R-centered physiological refinements, and lightweight domain adaptation. The framework was trained on the QT Database (QTDB) and externally evaluated on the Lobachevsky University Database (LUDB). The proposed method achieved a Macro F1-score of 0.9122 on QTDB, while obtaining 0.8236 under direct cross-dataset evaluation on LUDB. A lightweight adaptation strategy using 10% of LUDB records further improved the external Macro F1-score to 0.8451. These results demonstrate that progressively refining R-centered physiological guidance and input representations, together with lightweight adaptation, improves ECG wave delineation and cross-dataset generalization.

CCS Concepts: • **Computing methodologies → Machine learning**; • **Applied computing** → *Life and medical sciences.*

Additional Key Words and Phrases: Electrocardiography, Deep learning, CNN-BiLSTM, QTDB, LUDB, ECG and Wave Delineation, Cardiac signal processing

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1 Introduction

1.1 Motivation

Electrocardiography (ECG) is one of the most widely used non-invasive diagnostic tools for assessing cardiac electrical activity and plays a central role in the diagnosis of cardiovascular diseases [1]. Beyond heart rate estimation and rhythm

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analysis, accurate delineation of the P wave, QRS complex, and T wave is essential for measuring clinically significant intervals such as the PR, QRS, and QT intervals, which aid in the diagnosis of conduction abnormalities, ischemia, and various cardiac arrhythmias [2]. Consequently, automated ECG wave delineation has become a fundamental component of modern computer-aided cardiac diagnosis systems.

1.2 Existing Methods and Current Challenges

Early automated ECG delineation methods primarily relied on signal-processing techniques, including derivative-based, wavelet-based, and rule-based approaches for waveform boundary detection [3, 4]. Among these, the Pan–Tompkins algorithm remains a widely adopted method for robust QRS complex detection and continues to serve as a reliable foundation for heartbeat localization [3]. More recently, deep learning methods based on CNNs, encoder–decoder architectures, U-Net variants, BiLSTM networks, and attention-based models have substantially improved ECG delineation performance on benchmark datasets such as QTDB and LUDB [5–7]. Recent studies have also explored beat-wise segmentation, knowledge-guided learning, and R-peak-informed processing to improve waveform localization and accuracy across varying ECG morphologies [1, 8, 9].

Despite these advances, accurate delineation of P and T wave boundaries remains challenging due to their low amplitudes, morphological variability, noise sensitivity, and reduced generalization across different datasets [5, 10]. Furthermore, although recent studies have incorporated physiological priors into deep learning frameworks, the extent to which physiological guidance should influence the learning process remains insufficiently explored [1, 9]. Excessively rigid physiological constraints may limit model flexibility, whereas unconstrained models may fail to exploit cardiac temporal relationships.

1.3 Research Gap

While recent deep learning models have substantially improved ECG delineation accuracy, existing approaches primarily focus on designing increasingly sophisticated network architectures or incorporating physiological priors to enhance waveform localization [1, 2, 9, 11]. However, lesser attention has been paid to understanding how physiological guidance should be integrated into the learning process. In particular, it remains unclear whether imposing stronger physiological constraints consistently improves delineation performance or whether excessive guidance can inadvertently reduce model flexibility and generalization. This question is especially important for the delineation of low-amplitude P and T waves, where subtle morphological variations and inter-patient differences present significant challenges [5, 10]. Motivated by this observation, we systematically investigate the role of R-centered physiological guidance through a sequence of refined models, aiming to identify a balance between physiological consistency and data-driven learning.

1.4 Contributions

The main contributions of this work are summarized as follows:

- A physiologically guided R-centered CNN–BiLSTM framework is proposed for automated ECG wave delineation, integrating temporal cardiac context into the learning process for accurate sample-wise waveform segmentation.
- The effects of progressively introducing physiological guidance through six experimental configurations (R1–R6) are systematically investigated, providing a comprehensive analysis of the trade-off between physiological constraints and data-driven learning.

- The results demonstrate that appropriately designed physiological guidance improves cross-dataset generalization while maintaining competitive delineation performance.
- The proposed framework is validated on the QT Database (QTDB) and further evaluated for generalization on the Lobachevsky University Database (LUDB) using waveform classification and boundary localization metrics.

2 Related Work

2.1 Classical ECG Delineation

Early ECG delineation methods predominantly relied on signal-processing techniques, including derivative-based filtering, adaptive thresholding, wavelet transforms, and rule-based decision algorithms [3, 4]. The Pan–Tompkins algorithm established a reliable framework for real-time QRS detection and remains one of the most influential approaches for heartbeat localization [3]. Subsequent studies extended these principles to P and T wave delineation using adaptive signal-processing techniques and morphology-based representations, providing interpretable and computationally efficient solutions [4, 12]. However, these approaches usually require handcrafted features and exhibit reduced accuracy under noise, morphological variability, and pathological ECG patterns [4, 12].

2.2 Deep Learning-based ECG Delineation

The emergence of deep learning has advanced automated ECG delineation by enabling hierarchical feature learning directly from raw ECG signals [1, 2]. Deep learning approaches based on CNNs, encoder–decoder architectures, BiLSTM networks, and attention mechanisms have demonstrated strong performance on benchmark datasets such as QTDB and LUDB[2, 6, 7, 10, 11]. Representative approaches, including ECG_SegNet, knowledge-based encoder–decoder models, Peak Attention U-Net, and recent segmentation frameworks for diverse arrhythmias, have reported highly accurate waveform classification and boundary localization [1, 2, 10, 11]. Nevertheless, existing studies primarily focus on improving network architectures while treating physiological information as prior knowledge rather than systematically investigating its influence on model behavior.

2.3 Physiological Guidance and Positioning of Our Work

Recent studies have begun incorporating physiological knowledge through R-peak priors, beat-centric processing, adaptive windowing, and temporal constraints to improve ECG delineation [1, 8, 9, 13]. These methods demonstrate that physiological information can enhance waveform localization, particularly under challenging signal conditions [1, 9, 10]. However, the relative contribution of physiological guidance and the extent to which it should constrain deep learning models remain insufficiently understood. In contrast to proposing another network architecture, this work investigates the integration of R-centered physiological guidance to understand impact on delineation accuracy and cross-dataset generalization.

3 Methodology

3.1 Problem Formulation

ECG delineation is formulated as a sequence labeling problem in which each ECG sample is assigned to one of four classes: Background, P wave, QRS complex, or T wave. Given an input ECG segment

$$X = \{x_1, x_2, \dots, x_N\},$$

where N denotes the number of samples, the objective is to learn a mapping

$$f : X \rightarrow Y,$$

where

$$Y = \{y_1, y_2, \ldots, y_N\}, \quad y_i \in \{0, 1, 2, 3\},$$

representing the four waveform classes. The model predicts sample-wise waveform labels, enabling continuous ECG delineation. Model parameters are optimized using supervised learning by minimizing the classification loss between predicted labels and expert annotations.

The expert annotations were converted into sample-wise masks using the waveform interval annotations. For QTDB, the `pu0` annotations were interpreted by pairing the opening and closing interval markers; intervals associated with `p`, `N`, and `t` were assigned to the P-wave, QRS-complex, and T-wave regions, respectively, with all remaining samples assigned to background. The resulting sample-wise masks were aligned with the corresponding ECG samples before window extraction.

3.2 Baseline Framework

The baseline model consists of a hybrid CNN–BiLSTM architecture designed for sample-wise ECG waveform delineation. Before model training, ECG signals are normalized and segmented into fixed-length R-centered windows with corresponding sample-level annotations for the four waveform classes. Each detected R peak defines a 360-sample input segment containing 120 samples before the R peak and 240 samples after it, placing the R peak at sample 120 of the input window.

Windows were generated independently around consecutive detected R peaks rather than using a fixed-stride sliding-window scheme. Consequently, no fixed overlap percentage was imposed; overlap between adjacent windows, when present, was determined by the temporal separation between consecutive R peaks. Candidate windows extending beyond the beginning or end of a record were discarded rather than padded.

Each input segment is processed by stacked one-dimensional convolutional layers that extract local morphological features. The extracted features are then processed by a bidirectional long short-term memory (BiLSTM) network to capture temporal dependencies between waveform components. Finally, a fully connected layer predicts the waveform label for each sample, producing continuous ECG delineation.

This CNN–BiLSTM architecture (Fig. 1) serves as the baseline throughout the study and provides the reference model upon which all subsequent physiological guidance strategies are evaluated.

3.3 Failure Analysis

Before incorporating physiological guidance, multiple training strategies were investigated to establish a reliable baseline. The initial baseline **R1** employed the standard cross-entropy loss and achieved competitive performance on both the QTDB validation set and the LUDB external dataset. To improve the recognition of relatively underrepresented waveform classes, weighted focal loss was introduced in **R2**. Although this increased the recall of P and T waves, it substantially reduced precision and degraded the overall accuracy, indicating that excessive emphasis on minority classes negatively affected model calibration.

Based on these observations, the weighting strategy was removed while retaining focal loss, resulting in the **R3** model. This configuration achieved the best balance between precision and recall and consistently outperformed

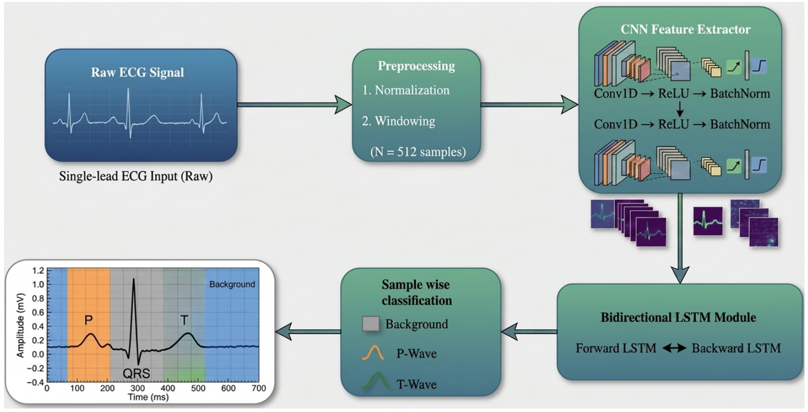

Fig. 1. Proposed CNN–BiLSTM baseline architecture for ECG wave delineation.

the previous variants on both internal validation and external testing. Hence, **R3** was selected as the reference baseline for all subsequent experiments. Having established a stable baseline, the remainder of this work investigates whether progressively incorporating physiological guidance further improves delineation performance and cross-dataset generalization.

3.4 Proposed R-Guided Framework

While the baseline CNN–BiLSTM model effectively learns morphological and temporal representations directly from ECG signals, it does not explicitly exploit the physiological organization of the cardiac cycle. In particular, the relative temporal relationship between the P wave, QRS complex, and T wave is governed by established cardiac electrophysiology, with the R-peak serving as a stable reference point across consecutive heartbeats. Motivated by this observation, we propose an R-guided framework that progressively incorporates physiological information into the delineation process while preserving the flexibility of data-driven learning.

Rather than imposing fixed physiological constraints from the outset, the proposed framework introduces physiological guidance through a sequence of refined models. An initial R-guided strategy explored strict temporal constraints around the detected R-peak. However, preliminary experiments demonstrated that excessive physiological restriction reduced the model's ability to adapt to natural waveform variability. Subsequently, the framework introduced an R-centered auditing mechanism with an expanded post-R temporal context, improving T wave characterization. Finally, a normalized temporal channel and limited cross-dataset adaptation were incorporated to provide additional physiological context while maintaining model flexibility. This progressive design (Fig. 2) enables the framework to leverage clinically meaningful temporal relationships without over-constraining the learning process, improving robustness across datasets.

3.5 Training Procedure

The proposed framework was trained using expert-annotated ECG signals from the QTDB [6, 14]. ECG recordings were segmented into fixed-length windows with corresponding sample-wise annotations. Model optimization was performed using supervised learning with hyperparameters selected based on validation performance. To assess generalization, the trained model was evaluated on the independent LUDB without retraining [7, 14]. For the final experimental

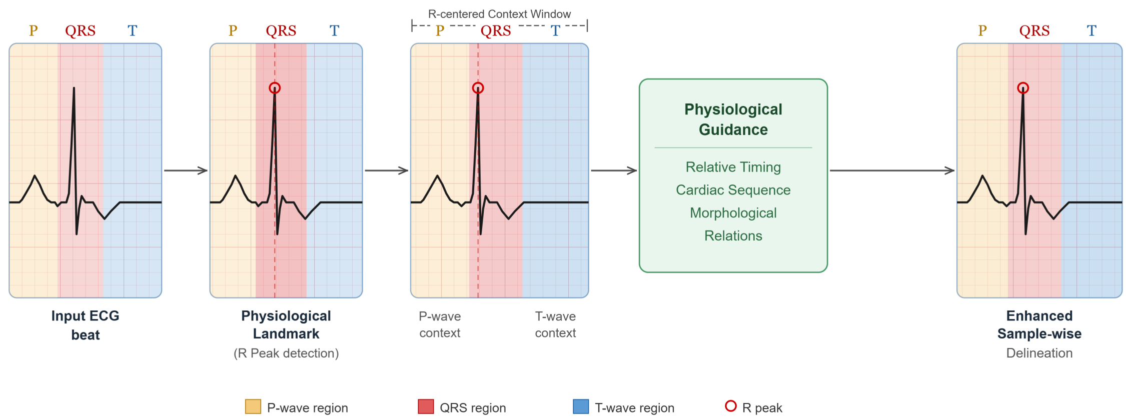

Fig. 2. Proposed R-guided framework for ECG wave delineation.

configuration, approximately 10% of the LUDB training records were used for lightweight adaptation to evaluate whether minimal domain-specific information improves cross-dataset performance while preserving generalization from QTDB, as in Table 1.

Before model training, the input ECG windows were standardized using the mean and standard deviation computed from the QTDB training windows only. Specifically, each input sample was normalized as $\hat{x} = (x - \mu_{\text{train}})/\sigma_{\text{train}}$, where μ_{train} and σ_{train} denote the global mean and standard deviation of the QTDB training windows. The same training-set statistics were applied to the QTDB validation data and to LUDB data during external evaluation and adaptation, without recomputing normalization statistics from either validation or LUDB samples.

4 Experimental Evaluation

4.1 Datasets

Experiments were conducted using two publicly available benchmark datasets on Physionet[14] for ECG waveform delineation, as summarised in Table 1. The QTDB was used for model development, training, and validation because it provides expert annotations for P waves, QRS complexes, and T waves[6]. A total of 105 QTDB records were used, corresponding to all records available in the QTDB directory used in our experiments. For each QTDB record, the first recorded signal channel was selected for processing. ECG recordings were converted into fixed-length windows with sample-wise labels corresponding to the four delineation classes: Background, P wave, QRS complex, and T wave.

Across the resulting QTDB R-centered windows, the sample-wise label distribution contained 25,775,702 background samples, 4,830,282 P-wave samples, and 8,961,616 T-wave samples, corresponding to 65.14%, 12.21%, and 22.65% of the labeled samples, respectively.

To evaluate the generalization capability of the proposed framework, experiments were performed on the Lobachevsky University Electrocardiography Database (LUDB), which contains independently annotated ECG recordings exhibiting broader morphological variability [7]. All 200 LUDB records available in the LUDB directory used in our experiments were processed. For each LUDB record, Lead II (`ii`) was selected for processing.

QTDB recordings were retained at their native sampling frequencies without resampling. For LUDB, the selected Lead II signal was downsampled from 500 Hz to 250 Hz using **polyphase resampling** with an upsampling factor of **1** and a downsampling factor of **2**. The corresponding annotation sample locations were scaled by the same factor to maintain temporal alignment with the resampled signal.

Unless otherwise stated, models trained on QTDB were directly evaluated on LUDB without retraining to assess cross-dataset robustness. For the final configuration, approximately 10% of LUDB training records were used for lightweight adaptation to evaluate whether limited domain-specific information improves external performance.

Table 1. Datasets used for the training of the architecture

Dataset/Partition	Records	R-centered windows	Evaluation Protocol
QTDB - Training	84	87,111	Used in all experiments (R1–R6)
QTDB - Validation	21	22,799	Model selection and validation
LUDB - External evaluation	200	1,904	Used for cross-dataset testing (R1–R5)
LUDB (10%) - Domain Adaptation	20		Fine-tuning only in R6
LUDB - Held-out test	180		Held-out adapted-test set for R6

4.2 Evaluation Metrics

The performance of each model was evaluated using classification metrics computed at the sample level. Precision (P), Recall (R), and F1-score were calculated for each waveform class (Background, P wave, QRS complex, and T wave). Overall accuracy and macro-averaged F1-score were additionally reported to assess performance across classes, particularly the underrepresented P wave class. Since the primary objective of this work is accurate waveform delineation rather than rhythm classification, the F1-score was considered the principal evaluation metric throughout the experimental analysis. To assess robustness, all model variants were evaluated on both the QTDB validation set and the independent LUDB dataset using similar evaluation protocols.

4.3 Progressive Experimental Evaluation

4.3.1 Baseline Optimization (R1 - R3). We first established a baseline by evaluating three training strategies with identical network architectures. The initial model (R1) employed the standard cross-entropy loss and achieved strong delineation performance on both the QTDB validation set and the independent LUDB dataset. To improve the recognition of relatively underrepresented waveform classes, R2 replaced the objective with weighted focal loss. While this modification increased the recall of P and T waves, it also introduced a noticeable reduction in precision and overall accuracy, indicating that excessive class weighting shifted the model toward over-predicting minority classes.

To overcome this limitation, R3 adopted an unweighted focal loss while preserving existing training configuration. This resulted in a balanced trade-off between precision and recall, producing the highest overall performance among the baseline variants on both the QTDB validation set, as in Fig. 3 and Table 2. R3 was hence selected as the reference model for all subsequent physiological guidance experiments.

Table 2. Experimental results comparing Macro F1 scores across loss functions on QTDB and LUDB.

Experiment	Loss Function	QTDB Macro F1	LUDB Macro F1
R1	CrossEntropy	0.9026	0.8203
R2	Weighted Focal Loss	0.8420	0.7604
R3	Unweighted Focal Loss	0.9061	0.8236

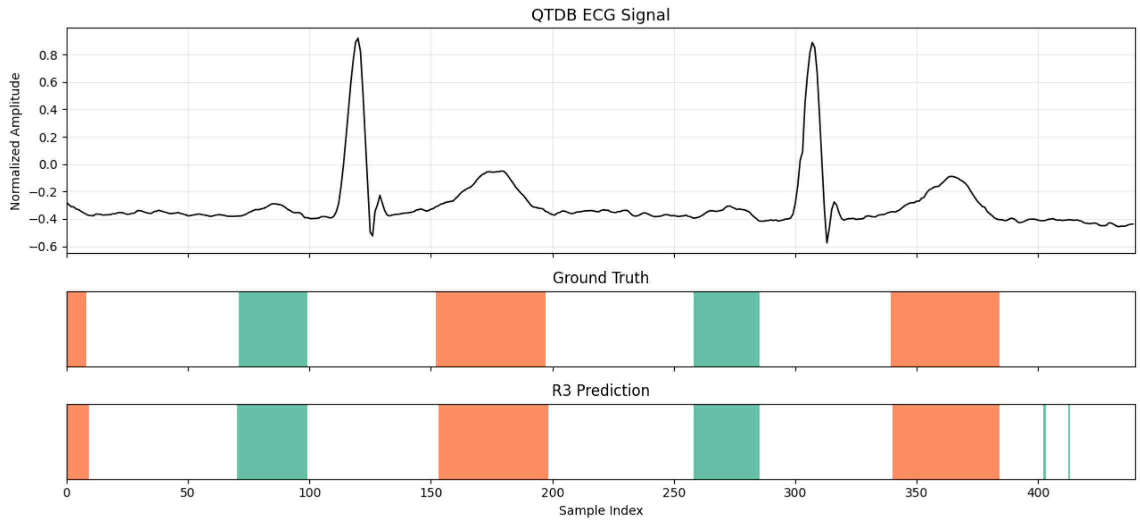

Fig. 3. Representative ECG delineation results on a QTDB validation beat. The top panel shows the ECG waveform, while the middle and bottom panels compare the ground-truth annotations with the predictions of the R3 baseline model.

4.3.2 Physiological Guidance. Having established R3 as the reference baseline, we investigated whether incorporating physiological information could further improve delineation performance, as in Table 3. The initial physiology-guided configuration (R4) applied an R-aware physiological decoder to the frozen predictions of the baseline model (R1), without retraining the neural network. By enforcing hard temporal constraints during decoding, this experiment isolated the effect of deterministic physiological guidance. This strategy produced a substantial reduction in performance on both QTDB and LUDB. As the underlying neural network remained unchanged, the observed degradation cannot be attributed to optimization or convergence. Instead, it indicates that rigid decoding discarded valid network predictions and reduced the model's ability to accommodate natural ECG variability, particularly for low-amplitude P and T waves.

Motivated by this observation, the subsequent configuration (R5) refined the R-centered input representation by introducing an R-detector audit together with an expanded post-R temporal context (POST320), while retaining the baseline training procedure. This modification improved T wave delineation while restoring the overall balance between precision and recall. Building upon this framework, the final model incorporated an additional normalized temporal channel and a limited adaptation strategy using approximately 10% of the LUDB training records. This configuration achieved the strongest cross-dataset performance, demonstrating that moderate physiological guidance can improve robustness and generalization without losing features of data-driven feature learning.

Table 3. Effect of progressively incorporating physiological guidance into the delineation framework. The R-aware decoder (R4) degraded performance, whereas the refined R Audit + POST320 strategy (R5) restored performance and improved over the baseline.

Experiment	Physiological Guidance	QTDB Macro F1	LUDB Macro F1
R3	Baseline (Unweighted Focal)	0.9061	0.8236
R4	Frozen Baseline + R-aware Decoder	0.7328	0.7119
R5	R Audit + POST320	0.9071	0.8266

4.4 Cross-dataset Generalization

To evaluate the accuracy of the proposed framework beyond the training distribution, all models were assessed on the independent LUDB dataset. While the baseline model (R3) demonstrated strong generalization despite being trained exclusively on QTDB, a measurable performance gap remained between internal validation and external testing due to differences in waveform morphology and dataset characteristics. The final framework addressed this limitation by incorporating a normalized temporal channel together with a limited adaptation strategy using approximately 10% of the LUDB training records. This resulted in improved delineation performance on the external dataset while preserving competitive performance on QTDB, demonstrating that modest domain-specific adaptation can effectively enhance cross-dataset robustness without requiring large-scale retraining, as shown in Table 4 and Fig. 4.

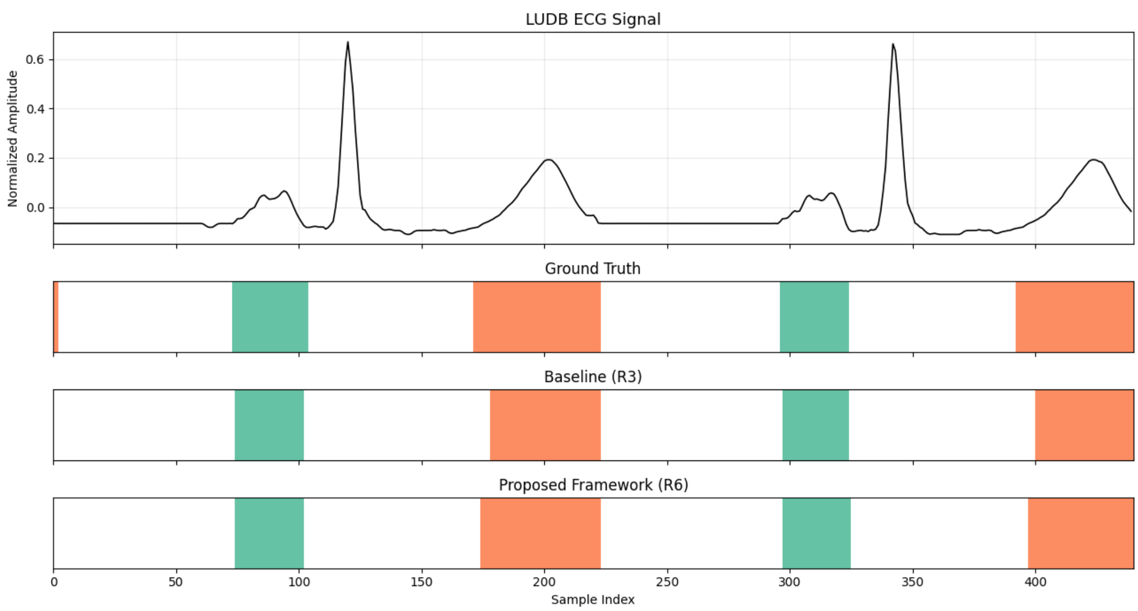

Fig. 4. Representative ECG delineation results of R6 on a held-out LUDB beat. The top panel shows the ECG waveform, while the middle and bottom panels compare the ground-truth annotations with the predictions of the proposed model.

Table 4. Comparison between pure cross-dataset generalization (R3) and lightweight domain adaptation (R6). These experiments represent different evaluation protocols and are therefore interpreted separately.

Experiment	Training Protocol	QTDB Macro F1	LUDB Macro F1
R3	QTDB ⟶ LUDB (Generalization)	0.9061	0.8236
R6	QTDB + 10% LUDB ⟶ Remaining LUDB (Adaptation)	0.9122	0.8451

4.5 Boundary-level analysis

Quantitative metrics provide an overall assessment of delineation performance; however, they do not fully capture differences in wave boundary localization. An inspection revealed that the baseline model occasionally exhibited imprecise localization of low-amplitude P and T waves. Introducing rigid physiological constraints did not consistently improve these predictions and, in several cases, reduced the model's ability to accommodate natural morphological variability. In contrast, the refined physiological framework produced smoother and more physiologically consistent waveform boundaries while preserving accurate QRS localization. These observations support the quantitative results and suggest that moderate physiological guidance, rather than rigid constraints, provides the most effective balance between physiological consistency and data-driven learning.

5 Discussion

The experimental results demonstrate that improving ECG delineation performance is not solely a matter of increasing model complexity or introducing additional physiological constraints. The experimental progression showed that rigid physiological constraints reduced the flexibility required to accommodate natural ECG variability, whereas progressively refined physiological guidance improved robustness while maintaining strong delineation accuracy.

The proposed framework consistently achieved competitive performance on the QTDB benchmark while demonstrating improved generalization to the independent LUDB dataset. These findings suggest that moderate physiological guidance provides a more effective balance between physiological consistency and data-driven feature learning than either the unguided baseline or rigid physiological constraints. This observation is particularly relevant for low-amplitude waveform components such as the P and T waves, where accurate boundary localization remains challenging.

Despite these promising results, limitations remain. The proposed framework was evaluated on two publicly available annotated datasets, and broader validation across additional clinical datasets would further establish its robustness. Also, the current study focuses on waveform delineation rather than downstream diagnostic tasks. Future work will investigate the integration of the proposed delineation framework into complete computer-aided ECG systems and evaluate its performance under a wider range of pathological conditions.

6 Conclusion

This paper presented an R-guided deep learning framework for automated ECG wave delineation and systematically investigated the influence of physiological guidance on segmentation performance. Through a sequence of progressively refined experimental configurations, we demonstrated that carefully designed physiological guidance improves cross-dataset robustness while preserving competitive delineation accuracy. Experimental evaluation on the QTDB and LUDB benchmark datasets showed that moderate physiological guidance provides a better balance between physiological consistency and model flexibility than rigid constraint-based approaches. These findings highlight the value of integrating physiological knowledge into deep learning frameworks for robust ECG delineation.

7 Ethics and Disclosures

LLMs have been used for grammar validation and spell checking, and generated content has been validated.

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